How to...

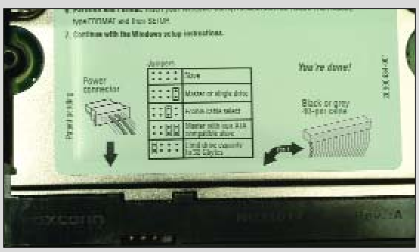
Upgrade your Hard Disk

Don't panic if you run out of hard disk space. Most computers can support two or more storage units, so expanding your storage space is easy. In this workshop, we will help you make the switch painless.

To use a new hard disk, you need to perform three tasks. First, physically install the hard disk and connect the power and data cables. Then, partition the hard disk to define data storage areas, and finally, format the partitions to allow the operating system to read from, and write to the hard disk. Before you begin the installation procedure, confirm that you have met the conditions outlined in the box "Pre-Install Checklist".

Install the hard disk

Open your computer cabinet to allow access to the components of your PC. Insert



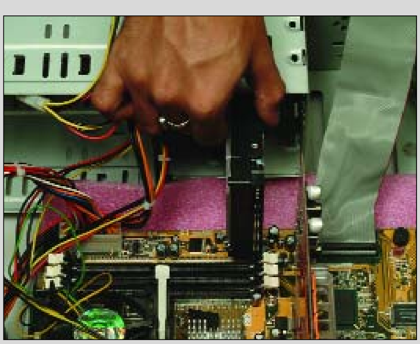
Set the jumper on the secondary drive

the hard drive into the 3.25-inch bay near the front of the case. Check for smooth runners along the bays, which will let you slide the drive in with ease. The hard disk's screw holes should be visible along the slit on the side. The power and data sockets should also be easily accessible.

Drive the screws in their respective sockets, and make sure the drive can't move. Fasten screws on both the sides of the hard drive bay.

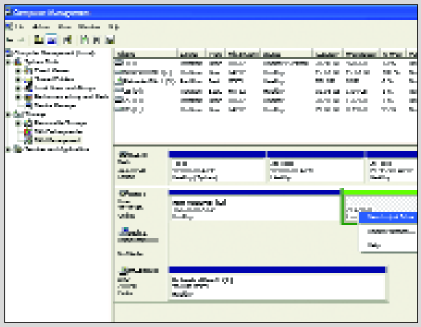
Make sure that you do not drop any screws inside the cabinet on the motherboard, and if you, pick them up before you turn on the power.

Connect the hard disk data cable to the free IDE connector on the motherboard.



Install the hard drive in the bay of the cabinet

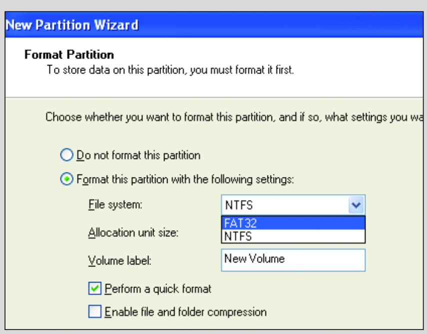
Connect the other end of the primary IDE cable to the back of the hard drive (*see "Cables and Jumpers" in box "Pre-Install*



The Disk Management tool lets you create an extended partition

Checklist"). This is very easy, and the red stripe running along the edge of the cable acts as a visual aid in doing this—the stripe should be on the edge nearer to the power socket. There is a notch on the connector, which also makes sure that you can only connect the drive the right way. Make sure that you check for this notch, and insert the cable carefully.

Now connect the power cable to the back of the hard drive. Notice the connector is 'D' shaped-it only plugs in one way, so you shouldn't need to force it. Once the power is plugged in, you are done!



Use the New Partition Wizard to format the partition

If you are performing a master-slave operation, here is what you need to do:

The above steps still hold good, but you have to connect the middle connector of the Primary IDE channel cable to the new hard drive. Attach the power cable, and you're all done.

Create partitions and format

Verify that they are no loose components or tools within the cabinet, replace all opened panels, and close the cabinet. Depending on your operating system, the steps that follow will vary. We will show you how to perform the remaining tasks for either Windows 98 or for Windows XP.

On booting up, the OS recognises the new hard drive. However, it's unformatted and not partitioned, so there's more to do.

Go to **Start > Control Panel > Administrative Tools > Computer Manage-ment** and click on Disk Management for information about the drives connected to the computer.

Right-click on the disk marked 'Unallocated', and click on the 'New partition' option to start the wizard. The first time, choose 'Primary partition' and then click Next.

Specify the amount of disk space that you want to use out of the maximum size shown. Choose NTFS or FAT32 as the file system. If you run Windows XP, and do not intend to use this PC with any other OS, choose NTFS for better reliability. Check the 'Perform a quick format' option to speed up this step.

Restart the computer and boot into Windows XP. Your new hard drive is ready-to-use.

Pre-install Checklist

1. Backup all critical data.
2. You should have a new hard disk, an IDE data cable (looks like a long flat ribbon), four mounting screws and small Philips head (+) screwdriver.
3. Check that the power to the computer is switched off from the switchboard. Ground yourself by touching a metal plate, or the side of the cabinet. This takes care of any static electricity issues during hard drive installation process.

Cables and Jumpers

If you do have a primary master hard drive, with an optical drive connected as the primary slave, don't change this arrangement. This reduces the connectivity fuss considerably. Since all new drives come pre-configured to be Master drives, you need not change the jumper settings. Just connect the drive as the secondary master on the secondary IDE channel.

If your secondary channel is in use, connect the drive as the slave on the primary channel. For that, follow the instructions on the hard drive itself, to change the jumper setting. Commonly used hard drives such as Samsung and Seagate don't need jumpers, if you are connecting the drive as a slave drive.

Speed Tip

Most new hard disks follow a fast data speed standard called ATA-100. To enable this, use special ATA-100 cables, or your speedy disk will crawl. ATA-100 cables are easily distinguishable. Since they have 80 data wires running on the cable, they look much finer in comparison to the 40 wires on the slower ATA-66 cables.